TEST CODE **000592**

FORM TP 21115

MAY/JUNE 2001

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

2 hours 40 minutes

01 JUNE 2001 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. **Answer ALL questions in Section I, and any TWO in Section II.**
2. **Begin the answer for each question on a new page.**
3. **Full marks may not be awarded unless full working or explanation is shown with the answer.**
4. **Mathematical tables, formulae and graph paper are provided.**
5. **Silent electronic calculators may be used for this paper.**
6. **You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

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SECTION I

Answer ALL the questions in this section.

ALL working must be clearly shown.

1. (a) Express in fractional form, in its lowest terms, the exact value of

$$\frac{3\frac{1}{7} - \frac{2}{3}}{2\frac{6}{7}}.$$

(3 marks)

(b) Write 0.08909

(i) in standard form

(ii) correct to two significant figures. **(3 marks)**

(c) A Building Society offers a rate of 11% per annum simple interest. Beth-Ann deposited $24 000 in the Society for 15 months.

(i) Calculate the amount of money due to her at the end of this period.

The Building Society charged her a processing fee of $1 500.

(ii) Calculate this fee as a percentage of the money originally deposited.

(5 marks)

Total 11 marks

2. (a) Factorize completely:

(i) $4y^2 + y$

(ii) $6x^2 - 13x - 5$ **(3 marks)**

(b) Solve the inequality

$$\frac{-2}{3}x \geq 4.$$

(3 marks)

(c) A restaurant bill of $350 was paid using $5 notes and $50 notes. The total number of notes used was 16.

Let x represent the number of $5 notes.
Let y represent the number of $50 notes.

(i) Write TWO equations in x and y to represent the information given.

(ii) Hence, calculate the number of $5 notes and the number of $50 notes.

(6 marks)

Total 12 marks

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3. (a) Solve the equation

$$6x - 8(x + 3) = 10.$$ **(3 marks)**

(b) $A = \begin{pmatrix} 2 & 1 \\ -3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} x \\ 4 \end{pmatrix}$ and $C = \begin{pmatrix} 12 \\ x \end{pmatrix}$

given that $AB = C$, calculate the value of x. **(3 marks)**

(c) The Universal set $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$A = \{0, 1, 2, 7, 9\}$
$B = \{3, 4, 5\}$
$C = \{2\}$

(i) Draw a Venn Diagram to represent the above information.

(ii) List, using set notation, the members of the set $A' \cap B'$. **(6 marks)**

Total 12 marks

4. (a) **Using ruler and compasses only:**

(i) Construct triangle DEF with $EF = DF = 7.5$ cm and $DE = 5$ cm.

(ii) Construct the circle which lies within triangle DEF and touches each side of the triangle.

(iii) Measure and write down the radius of this circle.

[All construction lines must be clearly shown.] **(6 marks)**

(b) P is the point (4, 2), Q is the point (12, 5) and R is the point (1, 3). Calculate

(i) the length of PR

(ii) the gradient of PQ

(iii) the equation of the line passing through R and parallel to PQ.

(6 marks)

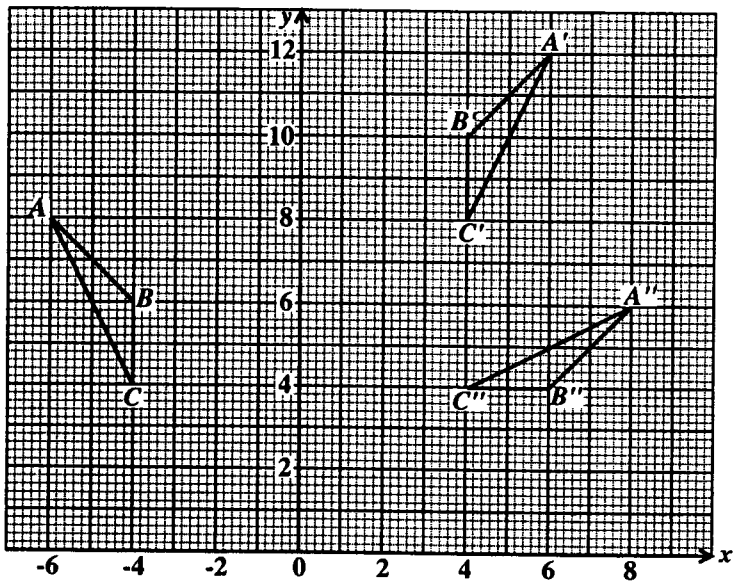
Total 12 marks

5. (a) A vertical pole stands on horizontal ground. From the top of the pole, h metres high, the angle of depression of a spot 10 m from the foot of the pole is 25°.

(i) Sketch a diagram to represent this information, showing the pole, the ground and the measurements given.

(ii) Calculate the value of h. **(6 marks)**

(b)



In the diagram above, triangle ABC is mapped onto triangles $A'B'C'$ and $A''B''C''$ under two different transformations. Describe FULLY the transformation which maps triangle ABC onto

(i) triangle $A'B'C'$

(ii) triangle $A''B''C''$. **(5 marks)**

Total 11 marks

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6. (a) Given that $g(x) = x + 3$ and $h(x) = x^2$,

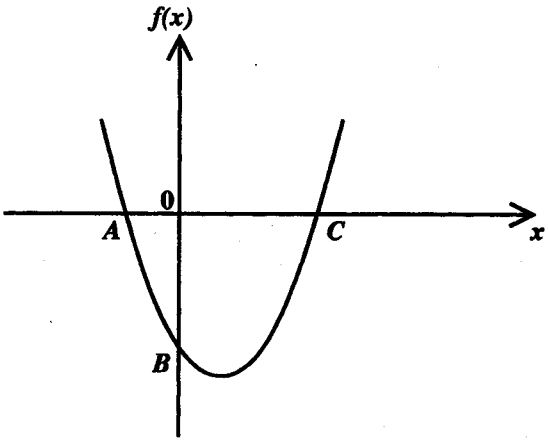
calculate

(i) $g(-5)$

(ii) $g^{-1}(7)$

(iii) $hg(0)$. **(5 marks)**

(b)



The diagram above shows the graph of the function $f(x) = x^2 - 3x - 10$.

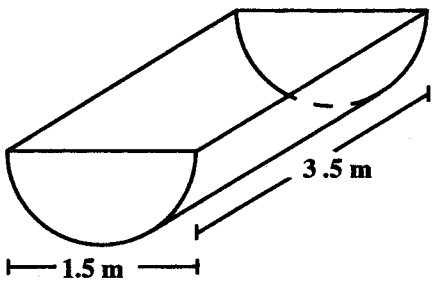
(i) Determine the values of x for which $f(x) = 0$.

(ii) What are the coordinates of the points A and C?

(iii) State the coordinates of B. **(6 marks)**

Total 11 marks

7.



[Take π = 3.142]

[Curved surface area of a cylinder = $2\pi r h$]

The diagram above, **not drawn to scale**, represents an open metal container. The cross-section of the container is a semi-circle of diameter 1.5 m. The length of the container is 3.5 m.

(a) Write down the radius of the cross-section of the container. **(1 mark)**

(b) Calculate, in m^2, to two decimal places

(i) the area of the cross-section of the container

(ii) the outer curved surface area of the container

(iii) the total outer area of the container. **(6 marks)**

(c) Calculate the capacity, in m^3, of the container. **(1 mark)**

(d) Water is poured into the container at a rate of 30 litres per minute.

Calculate the length of time, in minutes, it would take to just fill the container.

[$1m^3$ = 1000l] **(3 marks)**

Total 11 marks

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8. (a) The table below shows the income a company received from sales between 1995 and 2000.

Income ($)	50 000	60 000	65 000	40 000	30 000	60 000
Year	1995	1996	1997	1998	1999	2000

(i) Using a scale of 1 cm to represent $10 000 on the vertical axis and 2 cm to represent 1 year on the horizontal axis, draw a line graph to represent this information.

(ii) State the period during which the income showed the greatest rate of decrease.

(iii) State the period during which the income showed the greatest rate of increase.

(iv) Would you expect the sales in the year 2001 to increase or decrease? Using your graph state a reason for your answer. **(9 marks)**

(b) Determine the probability that in a year chosen at random, the income from sales would be greater than $60 000. **(1 mark)**

Total 10 marks

SECTION II

Answer TWO questions in this section.

RELATIONS AND FUNCTIONS

9. (a) Calculate the values of y for which

$5y^2 + 13y - 6 > 0$ **(5 marks)**

(b) (i) Write the expression $4x^2 - 9x + 1$ in the form

$$a(x + h)^2 + k,$$

where a, h and k are real numbers.

(ii) State whether the function

$$f(x) = 4x^2 - 9x + 1$$

has a maximum or minimum value.

(iii) Write down the value of x at which the maximum or minimum occurs.

(iv) Solve the equation

$$4x^2 = 9x - 1$$

giving your answers to two decimal points. **(10 marks)**

Total 15 marks

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10. (a) A farmer grows x acres of peas and y acres of tomatoes. He has 12 acres available to plant peas and tomatoes.

(i) Write an inequality in x and y to satisfy this condition.

The farmer must plant at least 2 acres of peas and at least 3 acres of tomatoes.

(ii) Write TWO inequalities to satisfy these conditions.

The number of acres of tomatoes planted must NOT be more than twice the number of acres of peas planted.

(iii) Write an inequality in x and y to satisfy this condition.

(iv) Using a scale of 1 cm to represent one acre on each axis, draw graphs of the inequalities you have written at (i), (ii) and (iii) above.

(v) By shading the UNWANTED region, label as S the region which satisfies all four inequalities. **(11 marks)**

(b) The farmer makes a profit of \$75 on each acre of peas and \$50 on each acre of tomatoes.

(i) Write an expression in x and y for the total profit, P, which may be earned on the peas and tomatoes.

(ii) Use the graph to determine the number of acres of peas and tomatoes the farmer should plant in order to make the maximum profit.

(iii) State the maximum profit. **(4 marks)**

Total 15 marks

GEOMETRY AND TRIGONOMETRY

11. (a) Q, R and S are three points on level ground. S is due east of Q and $QS = 45$ metres. The bearing of R from Q is 042° and the bearing of S from R is 110°.

(i) Draw a diagram to show this information indicating clearly the bearings and distances given.

(ii) Calculate the bearing of R from S.

(iii) Calculate, in metres, the distance RS, giving your answers correct to 2 significant figures.

[Accurate drawings for this question will not be accepted.]

(10 marks)

(b) Two towns C and D are situated at (35°N, x°W) and (35°N, 18°W) respectively. C is due west of D. The distance CD, measured along the parallel of latitude is 2100 km.

Calculate, to the nearest degree, the value of x.
[Use $\pi = 3.14$ and $R = 6370$ km] **(5 marks)**

Total 15 marks

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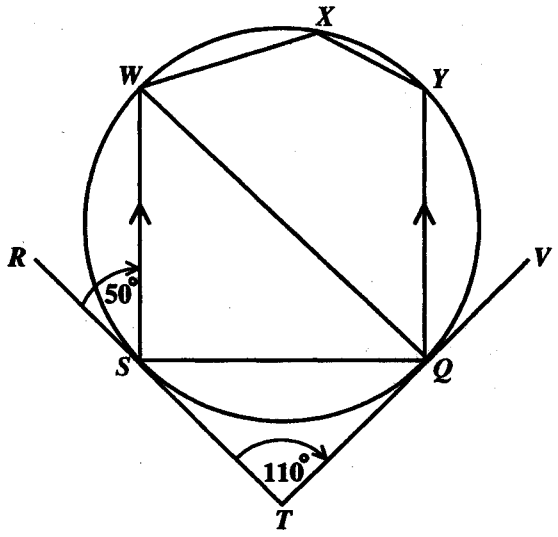
12. (a)



In the diagram above, **not drawn to scale**, *RST* and *TQV* are tangents to the circle *QSWXY*. Angle *RSW* = 50°, angle *STQ* = 110°, and *SW* is parallel to *QY*.

Calculate, giving reasons for your answers, the sizes of angles

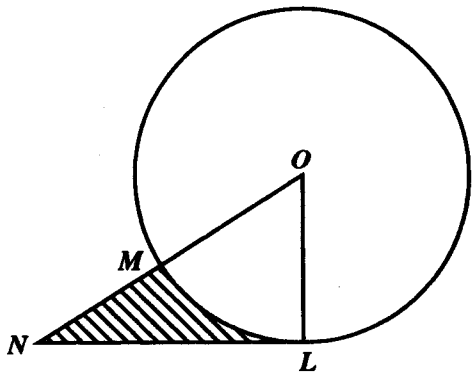
(i) *WQS*

(ii) *WSQ*

(iii) *WQY*

(iv) *WXY*. **(8 marks)**

(b)



The diagram above, **not drawn to scale**, shows a circle, centre *O*, radius 15 cm. The length of the minor arc *LM* is 9 cm and *LN* is a tangent to the circle. *OMN* is a straight line.

(i) Calculate, in radians, the size of angle *MOL*.

Hence, calculate

(ii) the area of the minor sector *OML*

(iii) the area of the shaded region. **(7 marks)**

Total 15 marks

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VECTORS AND MATRICES

13. (a) R is the matrix $\begin{pmatrix} 3r & 1 \\ s & 2s \end{pmatrix}$

(i) State, in terms of s and r, the determinant of R.

(ii) If $r = \frac{1}{3}$ and $s = 4$, determine the inverse of R.

(iii) State a pair of values of r and s, not including zero, which would make the matrix R a singular matrix. **(7 marks)**

(b) The position vectors of the points V, E and D relative to an origin O are

$$\overrightarrow{OV} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \overrightarrow{OE} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}, \overrightarrow{OD} = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

respectively.

(i) Express the following vectors in the form $\begin{pmatrix} a \\ b \end{pmatrix}$:
$\overrightarrow{VE}$, $\overrightarrow{DV}$, $\overrightarrow{ED}$.

(ii) Prove that the points V, E and D lie on a straight line and show their relative positions on the line.

(iii) State the value of the ratio $EV:DV$. **(8 marks)**

Total 15 marks

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14. (a) Q is the transformation represented by the matrix $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

R is the transformation represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(i) Describe FULLY the transformations represented by Q and by R.

(ii) Determine the SINGLE matrix, L, which represents the transformation R followed by Q.

(iii) Calculate the coordinates of the image of the point $(-1, 3)$ under L.

(iv) Determine L^{-1}, the inverse of L.

(v) Using L^{-1}, calculate the coordinates of the point whose image under L is $(-5, 2)$.

(8 marks)

(b) The matrix $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ maps $(1, 3)$ onto $(-1, 3)$.

(i) Determine the values of a, b, c and d.

(ii) The point $(1, 3)$ lies on a line V. Given that $(-4, 2)$ also lies on V, determine the equation of V', the image of V, under the transformation T. **(7 marks)**

Total 15 marks

END OF TEST